RHCSA Mock Exam — Questions & Solutions

EX200 | Node 1 + Node 2 + Extra Questions

NODE 1

Q1 — Network Configuration

hostname = node1.lab.example.com

ip-address = 172.25.250.10

subnetmask = 255.255.255.0

gateway = 172.25.250.254

dns = 172.25.254.254

Solution:

```
#nmcli connection show
#nmcli connection modify "Wired connection 1" ipv4.address 172.25.250.10/24
ipv4.gateway 172.25.250.254 ipv4.dns 172.25.254.254 ipv4.method manual
connection.autoconnect yes
# hostnamectl set-hostname node1.lab.example.com
#systemctl restart NetworkManager
#systemctl enable NetworkManager
#nmcli connection up "Wired connection 1"
Verify:
#reboot
#hostname
#cat /etc/resolv.conf
#ip a
#route -n
# ping 172.25.250.10
# ping 172.25.250.254
```

Q2 — Configure Repository

Configure YUM repository with the 2 given links (BaseOS and AppStream)

BaseOS: http://content.example.com/rhel8.0/x86_64/dvd/BaseOS

AppStream: http://content.example.com/rhel8.0/x86_64/dvd/AppStream

Solution:

```
# yum repolist
#vim /etc/yum.repos.d/local.repo
Inside file:
[BaseOS]
```

```
name=BaseOS
baseurl=http://content.example.com/rhel8.0/x86_64/dvd/BaseOS
enabled=1
gpgcheck=0
[AppStream]
name=Appstream
baseurl=http://content.example.com/rhel8.0/x86_64/dvd/AppStream
enabled=1
gpgcheck=0
#yum clean all
#yum update
Verify:
# yum repolist
#yum install nmap -y
```

Q3 — Debug SELinux

Web server running on non-standard port "82" is having issues serving content, Debug and fix the issues.

-The web server can serve all the existing HTML files from '/var/www/html', Don't make any changes to these files.

-Web service should automatically start at boot time.

- curl servera.lab.example.com:82

Solution:

```
#systemctl start httpd
#systemctl enable httpd
# semanage port -l | grep http
# semanage port -a -t http_port_t -p tcp 82
# semanage port -l | grep http
#firewall-cmd --permanent --add-port=82/tcp
#firewall-cmd --reload
Verify:
#systemctl restart httpd
#curl servera.lab.example.com:82
```

Q4 — Cron Job

-Configure a cron job that runs every 1 minutes and executes: logger "EX200 in progress" as the user natasha.

-Configure a cron job for user "natasha", cron must run daily at 2:23pm and inside executes the /usr/bin/echo "welcome"

Solution:

```
#systemctl start crond
#systemctl enable crond
```

```
#crontab -eu natasha
Inside crontab:
*/1 * * * * logger "EX200 in progress"
23 14 * * * /usr/bin/echo "welcome"
#systemctl restart crond
#systemctl enable crond
Verify:
#tail -f /var/log/messages | grep "EX200"
#crontab -lu natasha
```

Q5 — Create User Accounts with Supplementary Group

group: sysadms

-users: natasha harry sarah (sarah with nologin shell)

-natasha and harry should be the member of sysadms group.

-password for all users should be "trootent"

Solution:

```
#groupadd sysadms
#useradd -G sysadms natasha
#useradd -G sysadms harry
#which nologin
#useradd -s /usr/sbin/nologin sarah
#passwd natasha
#passwd harry
#passwd sarah
Verify:
#cat /etc/group (or cat /etc/group | grep sysadms)
#cat /etc/passwd
```

Q6 — Create a Collaborative DIR (change group owner, set the permissions along with sgid)

-Create the Directory "/home/sysadms" with the following characteristics.

-Group ownership of "/home/sysadms" should go to "sysadms" group.

-The directory should have full permission for all members of "sysadms" group but not to the other users except "root".

-Files created in future under "/home/sysadms" should get the same group ownership.

Solution:

```
#mkdir -p /home/sysadms
#chgrp sysadms /home/sysadms
#ls -ld /home/sysadms
#chmod 770 /home/sysadms
#chmod g+s /home/sysadms
```

```
Verify:
#su - harry
#touch /home/sysadms/test
#ls -ld /home/sysadms/test
```

Q7 — Configure NTP

Configure NTP chrony server is "classroom.example.com"

Solution:

```
#systemctl start chronyd
#systemctl enable chronyd
#timedatectl
#timedatectl set-ntp true
#timedatectl
#vim /etc/chrony.conf
Inside file add:
server classroom.example.com iburst
#systemctl restart chronyd
#systemctl enable chronyd
Verify:
#chronyc sources
```

Q8 — Configure AutoFS

NFS exports the /home/guests to your system.

-remoteuser6 home directory is classroom.example.com:/home/remoteuser6

-remoteuser6 home directory should be automounted locally beneath at /home/guests/remoteuser6

-while login with remoteuser6 then only home directory should accessible from your system; remoteuser6 password is "redhat"

Solution:

```
#systemctl start autofs
#systemctl enable autofs
#getent passwd remoteuser6
#vim /etc/auto.master
Inside file add:
/home /etc/auto.misc
#vim /etc/auto.misc
Inside file add:
remoteuser6 -rw,nfs,sync classroom.example.com:/home/remoteuser6
#systemctl restart autofs
#systemctl enable autofs
Optional: chmod 777 /home/
Verify:
```

```
#su - remoteuser6
#pwd
```

Q9 — Create User with UID

Create user 'bob' with 2112 uid and set the password 'trootent'

Solution:

```
#useradd -u 2112 bob
#passwd bob
Verify:
#id bob
```

Q10 — Create Archive

create an archive '/root/test.tar.gz' of '/var/tmp' dir and compress it with gzip.

Solution:

```
#tar -zcvf /root/test.tar.gz /var/tmp
Verify:
#ls
```

Note: gzip uses -zcvf (.gz), bzip2 uses -jcvf (.bz2)

Q11 — Find Files by Owner

locate all files owned by sarah and copy it under /root/find_user

Solution:

```
#mkdir /root/find_user
#find / -user sarah -type f
#find / -user sarah -type f -exec cp -rfvp {} /root/find_user \;
Verify:
#ls /root/find_user
```

Q12 — Grep and Redirect

Find a string 'home' from '/etc/passwd' and put it into '/root/search.txt' file

Solution:

```
#grep "home" /etc/passwd
#grep "home" /etc/passwd > /root/search.txt
Verify:
#cat /root/search.txt
```

Q13 — Create a Container Image Named 'monitor'

Create a container image named monitor with the link provided in the exam instructions. Do not edit the Containerfile.

Login with user "student" to perform all the questions related to container.

Solution:

```
as user:
#ssh student@node1
#podman login <registry-link-from-instructions>
User: <username>
Password: <password>
#wget <link-from-question>
#podman build -t monitor .
#podman images
```

Q14 — Create Container as systemd Service

1. Create a Container name asciipdf

2. Use monitor image for asciipdf which you previously created

3. Create a systemd service name container-asciipdf for "student" user only

4. Service will automatically start across reboot, do no any manual intervention.

5. Local host Directory /opt/files attach to Container directory /opt/incoming.

6. Local host Directory /opt/processed attach to container directory /opt/outgoing

(you don't have to install podman or container-tools, everything is pre-installed)

Solution:

```
As root:
#ssh root@node1
#mkdir /opt/files /opt/processed
#chown student:student /opt/files /opt/processed
#chmod 777 /opt/files /opt/processed
#loginctl enable-linger student
As user (student):
#podman run -d --name ascii2pdf -v /opt/files/:/opt/incoming/:Z -v
/opt/processed/:/opt/outgoing/:Z monitor
#podman ps
To verify container works:
#touch /opt/files/data
#ls /opt/processed
#file /opt/processed/data
#mkdir -p .config/systemd/user
#cd .config/systemd/user
#podman generate systemd --name ascii2pdf --new --files
#ls
#systemctl --user daemon-reload
#systemctl --user start container-ascii2pdf.service
#systemctl --user enable container-ascii2pdf.service
As root:
# reboot
As user:
```

```
#ssh student@node1
#podman ps
```

NODE 2

Q15 — Break Node 2 Root Password

break node 2 password (reset the root password for Node 2)

Solution:

```
1. Reboot Node 2, interrupt GRUB and select the kernel entry
2. Press Ctrl+e to edit the boot entry
3. Find the line starting with "linux" and append: rd.break at the end
4. Press Ctrl+x to boot
#mount -o remount,rw /sysroot
#chroot /sysroot
#passwd root
#touch /.autorelabel
#exit
#exit
(system will reboot automatically)
```

Q16 — Configure YUM Repository

Configure YUM repository (same as Q2 but with different links provided in the exam)

Solution:

```
NO ANSWER (same procedure as Q2, use the links given in the exam instructions)
```

Q17 — Swap Partition

Add a swap partition of 512MB and mount it permanently.

Solution:

```
#free -m
#lsblk
#fdisk /dev/vdb
-n (new partition)
-p (primary)
Enter (default first sector)
Enter (default or +512MiB for last sector)
+512MiB
-t (change type)
Enter
swap (or type the number for swap)
-w (write)
#partprobe /dev/vdb
#mkswap /dev/vdb1
```

```
(copy the UUID from the output)
#vim /etc/fstab
Inside file add:
UUID=<uuid-from-mkswap> swap swap defaults 0 0
#swapon -a
#free -m
Verify:
#reboot
#lsblk
```

Q18 — Create Logical Volume and Mount Permanently

-Create the logical volume with the name "lvname" by using 50 extents from the volume group "groupname".

-Consider size of volume group extent as "8MB".

-Mount it on '/mnt/lvname' with file system ext3.

Solution:

```
#lsblk
#fdisk /dev/vdb
-n (new partition)
-p (primary)
Enter
Enter
+1G
-t (change type)
lvm (or enter the number for LVM)
-w
#partprobe /dev/vdb
#pvcreate /dev/vdb2
#pvs
#vgcreate -s 8M groupname /dev/vdb2
#vgs
#lvcreate -l 50 -n lvname groupname
#mkfs.ext3 /dev/groupname/lvname
#mkdir /mnt/lvname
#vim /etc/fstab
Inside file add:
/dev/groupname/lvname /mnt/lvname ext3 defaults 0 0
#mount -a
#lsblk
Verify:
#reboot
#lsblk
```

Q19 — Resize Logical Volume

Resize the logical volume "lvname" to be 230MiB. After reboot size should be in between 200MB to 300MB.

Solution:

```
#lvextend -r -L 320MiB /dev/groupname/lvname
#mount -a
#resize2fs /dev/groupname/lvname
Note: if the file system is xfs use: xfs_growfs /dev/groupname/lvname
Verify:
#reboot
#lsblk
```

Q20 — Configure System Tuning

Choose the recommended 'tuned' profile for your system and set it as the default.

Solution:

```
#systemctl start tuned
#systemctl enable tuned
#tuned-adm active
#tuned-adm recommend
#tuned-adm profile <name-from-recommend-output>
#tuned-adm active
#systemctl restart tuned
#systemctl enable tuned
```

EXTRA QUESTIONS (one of these appears in the exam)

Extra Q1 — Create a Simple Script

Create a mysearch script to:

-locate all files under /usr/share having size less than 10M.

-after executing mysearch script, the searched files have to be copied under /root/myfiles

Solution:

```
#mkdir /root/myfiles
#vim mysearch.sh
Inside file:
#!/bin/bash
find /usr/share -type f -size -10M -exec cp -prvf {} /root/myfiles \;
#chmod u+x mysearch.sh
#./mysearch.sh
Verify:
#ls -a /root/myfiles
```

Extra Q2 — Expire Password

Make all the local users present in the system have their password expire in 30 days.

Solution:

```
For single user harry, 30 days:
#chage -M 30 harry
For all users, change the config file:
#vim /etc/login.defs
Search for PASS_MAX_DAYS and change the value:
PASS_MAX_DAYS 30
Verify:
#cat /etc/shadow
```

Extra Q3 — Configure Application Environment Variable

Configure the application RHCSA as a harry user; when login it will show the message "welcome to advantage pro"

Solution:

```
#su - harry
#vim .bash_profile
Inside file add:
RHCSA="welcome to advantage pro"
export RHCSA
echo $RHCSA
#source .bash_profile
Verify:
logout
# su - harry
(should display: welcome to advantage pro)
```

Extra Q4 — Sudo Privileges

Configure sudo for group 'elite' so that its members should have sudo access with no password.

Solution:

```
#visudo
Inside file add:
%elite ALL=(ALL) NOPASSWD: ALL
Verify:
Switch to a user from elite group (create one if needed and add to elite group)
#su - username
#sudo useradd test
#sudo cat /etc/passwd
```

Extra Q5 — ACL

Copy "/etc/fstab" file to "/var/tmp" then configure "/var/tmp/fstab" file permissions with ACL:

-the file /var/tmp/fstab should be owned by "root"

-the file /var/tmp/fstab should belong to the group "root"

-the file /var/tmp/fstab shouldn't be executable by anyone

-the user "sarah" should be able to read and write to the file

-the user "harry" can neither read nor write to the file

-other users (future and current) should be able to read /var/tmp/fstab

Solution:

```
#cp /etc/fstab /var/tmp
#chown root:root /var/tmp/fstab
#chmod ugo-x /var/tmp/fstab
#setfacl -m u:sarah:rw /var/tmp/fstab
#setfacl -m u:harry:0 /var/tmp/fstab
#chmod o=r /var/tmp/fstab
Verify:
#getfacl /var/tmp/fstab
```